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$$\int \frac{dx}{x^2 \sqrt{x^2 - 4}}$$

$$\text{Vorm } \sqrt{x^2 - a^2} \text{ met } a = 2, \text{ stel } x = \frac{2}{\cos t} \Rightarrow dx = \frac{2 \sin t}{\cos^2 t} dt$$

$$\sqrt{x^2 - 4} = \sqrt{\frac{4}{\cos^2 t} - 4} = \frac{2 \sin t}{\cos t}$$

$$= \int \frac{1}{\frac{4}{\cos^2 t} \cdot \frac{2 \sin t}{\cos t}} \cdot \frac{2 \sin t}{\cos^2 t} dt$$

$$= \int \frac{\cos^3 t}{8 \sin t} \cdot \frac{2 \sin t}{\cos^2 t} dt$$

$$= \int \frac{\cos t}{4} dt$$

$$= \frac{\sin t}{4} + c$$

$$\text{Terugsubstitutie: } \cos t = \frac{2}{x}, \sin t = \frac{\sqrt{x^2 - 4}}{x}$$

$$= \frac{\sqrt{x^2 - 4}}{4x} + c$$

References